Homework 2 Solutions

Answers are shown in red beneath each question.

Part I: Sampling and Experimental Design

(**1**) Define each of the following terms.

(a) Explanatory variable

The independent variable, the one we manipulate or group by in order to see how the response variable changes.

(b) Response variable

The outcome, or dependent, variable - the variable on which we make comparisons across different values of the explanatory variable.

(c) Experimental study

A study that assigns subjects to one or more treatments and then observes the outcome of the response variable.

(d) Observational study

A nonexperimental study in which the researcher simply observes the values of the response and explanatory variables for each subject, without assigning treatments.

(e) Survey

A type of nonexperimental study in which subjects are selected from a population for the purpose of making inferences about that population.

(f) Lurking (confounding) variable

A variable that is not among the explanatory or response variables being studied but is associated with both of them, and which can therefore create the appearance of a relationship between them, or hide a real one.

(g) In one or two sentences, explain why a statistical **association** between two variables is not by itself evidence that one **causes** the other. Give an example of a lurking variable that could produce an association without a causal link.

An association only tells us that two variables move together; it does not tell us why. In an observational study a lurking variable may be driving both, and without random assignment we cannot rule that out. For example, ice cream sales and drownings rise together, but neither causes the other - summer heat is the lurking variable that drives both.

(Any sensible example is acceptable, provided the proposed lurking variable is plausibly associated with **both** the explanatory and the response variable.)

(**2**) Name the sampling method used in each of the following situations.

(a) A man stands outside a grocery store handing out questionnaires that ask shoppers to evaluate their shopping experience and the quality of the service. He does not approach shoppers whose hands are full of groceries; instead he asks every shopper who has only a few items or who has both hands free.

Convenience sampling.

(b) A gardener has planted 10 rows of corn in a garden box, and each row holds eight corn plants. The gardener wants to know the average height of the corn plants in the box, so he randomly selects rows 1, 5, and 9 and measures the height of every plant in those rows.

Cluster sampling. The rows are the clusters, and every plant within a selected row is measured.

(c) A company wants to know how its employees feel about compensation and benefits. Over the next few weeks, the company asks 30 employees from each of its branch locations to fill out a survey about compensation.

Stratified random sampling. The branch locations are the strata, and a fixed number of employees is drawn from each.

(d) A polling company wants to know what proportion of the citizens of Moscow, Idaho, are voting for the Republican candidate in the mayoral race. The company records this information for every 5th voter who leaves the voting booth.

Systematic sampling.

(e) A lumber company wants to estimate the amount of usable lumber on a plot of land. The company randomly selects 100 trees and measures the diameter, height, and volume of each one in order to estimate its yield in board feet.

Simple random sampling.

(f) Explain the difference between **sampling error** and **nonsampling error**, and give one example of each.

Sampling error is the error that arises simply because we observed a sample rather than the whole population. Even a perfectly executed random sample will not match the population exactly, and a different random sample would have given a slightly different estimate. It shrinks as the sample size grows. Example: a well-run poll of 1,000 voters reports 52% support when the true population value is 54%.

Nonsampling error is every other source of error. It comes from how the study was carried out rather than from the act of sampling, and it does **not** shrink as the sample size grows. Examples: undercoverage, where part of the population has no chance of being selected; nonresponse; response bias from a leading question; and data entry mistakes.

(g) A university emails a satisfaction survey to all 11,000 of its students, and 600 reply. The students who reply are far more likely to be the ones with a strong complaint. Name the source of bias at work here, and explain why emailing 40,000 students instead of 11,000 would **not** fix the problem.

This is **nonresponse bias**. The sample is effectively a voluntary response sample, because only those motivated to reply are represented.

Increasing the number of students contacted does not help, because the same mechanism still decides who replies, and dissatisfied students remain over-represented among the respondents. A larger sample makes the estimate more **precise**, but it is precisely centred on the wrong value - a bigger sample does not remove bias. The fix is to raise the response rate, or to follow up on a random subsample of the nonrespondents.

(**3**) Name the experimental design used in each of the following situations. Choose from **completely randomized design**, **randomized block design**, and **matched pairs design**.

(a) A nutritionist wants to compare three breakfast cereals. She recruits 90 volunteers and uses a random number generator to assign 30 of them to each cereal.

Completely randomized design.

(b) A dermatologist testing a new ointment applies the ointment to one arm of each patient and a placebo cream to that patient's other arm, deciding at random which arm receives which.

Matched pairs design. Each patient serves as their own control; this is the special case of a randomized block design in which every block has size two.

(c) An agronomist testing four fertilizers suspects that soil drainage strongly affects yield. He sorts his 40 plots into a "well-drained" group and a "poorly-drained" group, then randomly assigns the four fertilizers within each group.

Randomized block design, with drainage as the blocking variable.

(d) A sleep researcher recruits 200 subjects, and a coin flip decides whether each subject receives a new sleep medication or an identical-looking pill that contains no medication.

Completely randomized design.

(e) For the study in part (d), identify the **experimental units**, the **factor**, the **levels** of that factor, the **treatments**, and the **control group**. Was this study **blinded**? What additional information would you need in order to know whether it was **double blinded**?

Experimental units (subjects): the 200 recruited people. **Factor:** the sleep medication. **Levels:** two - active medication, and no medication. **Treatments:** the new sleep medication, and the placebo pill. **Control group:** the subjects who receive the identical-looking placebo pill.

The study is **blinded**, because the placebo is identical in appearance and so subjects cannot tell which treatment they received. To know whether it is **double blinded**, we would also need to know whether the researchers who interact with the subjects and measure the outcome are themselves unaware of who received which treatment.

(f) A study finds that students who attend more review sessions earn higher exam scores, but the review sessions were optional. Explain what **random assignment** would have allowed the researchers to conclude that this study cannot, and explain why **random sampling** alone is not enough to establish cause.

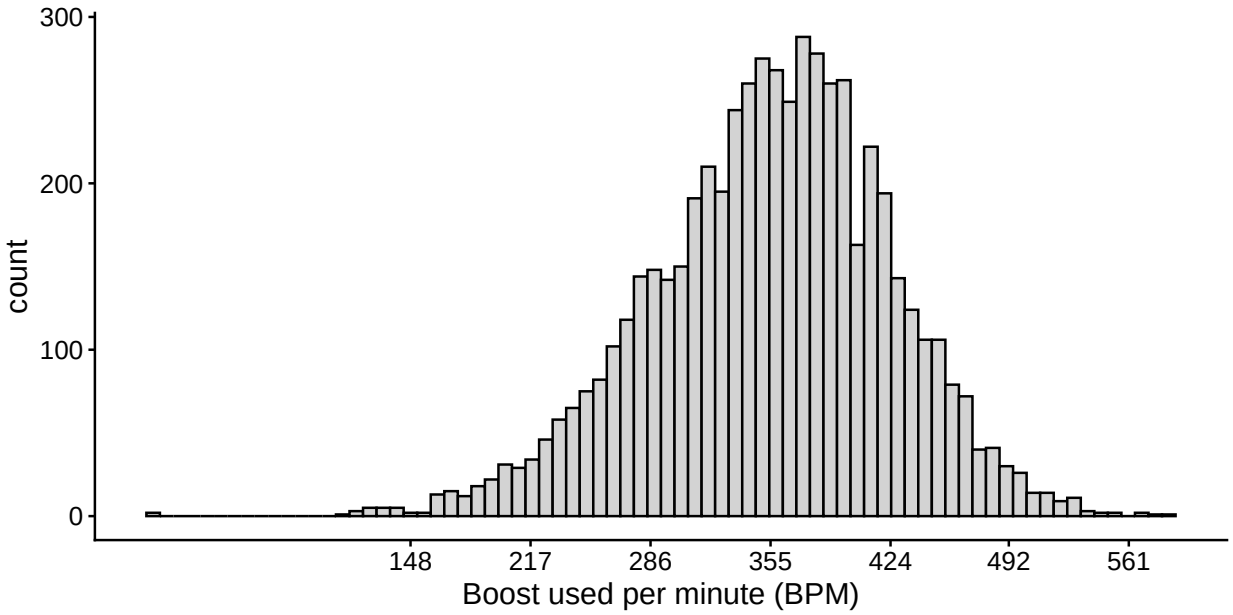
Because attendance was optional, the students who chose to attend may differ systematically from those who did not. They may be more motivated, or have more free time, and either would raise exam scores on its own. Motivation is a lurking variable, so this observational study can establish only an **association**.

Random assignment - deciding by chance which students attend - would balance those lurking variables across the two groups on average, so any difference in exam scores could be attributed to the review sessions themselves. That is what licenses a **causal** conclusion.

Random sampling does something different: it makes the sample representative, so the results **generalize** to the population. It says nothing about why the two groups differ, so it cannot establish cause on its own.

Part II: The Normal Distribution, z-scores, and Outliers

(**4**) Rocket League is a popular online video game and E-sport that emerged in 2015. Players from around the world compete in sports such as soccer (football), basketball, and hockey while controlling RC-like vehicles. The most popular game mode is doubles soccer, in which two teams of two players each try to outscore the other by hitting a soccer ball into the opposing goal. Each car carries a limited supply of **boost**, which is spent to accelerate and to fly and which is replenished by driving over boost pads on the field. A player's **boost used per minute (BPM)** measures how aggressively that player spends boost over the course of a match. The histogram below shows the distribution of BPM for $n = 5{,}714$ player-match records collected from ranked matches.



(a) Assume a mean of $\bar{x} = 355$ and a standard deviation of $s = 69$. In approximately what proportion of matches did a player use more than 424 BPM?

$$z = \frac{424 - 355}{69} = 1$$

So we want the proportion of observations greater than $\bar{x}+1s$. Under the empirical rule 68% of observations fall within $1s$ of the mean, leaving 32% split evenly between the two tails, so the upper tail is $13.5\% + 2.5\% = 16\%$.

(b) In approximately what proportion of matches was the BPM greater than 217 but less than 493?

$217 = \bar{x} - 2s$ and $493 = \bar{x} + 2s$, so by the empirical rule approximately 95%.

(c) How much boost per minute would a player need to use in a match in order to be in the top 2.5% of matches?

The empirical rule puts 95% of observations within $2s$ of the mean, so 2.5% sit above $\bar{x} + 2s$. The player would need at least $355 + 2(69) = 493$ BPM.

(d) In a new match, a player records a BPM of 562. Compute the z-score for this match and explain what that z-score means in relation to the average BPM. How many standard deviations from the mean is this performance?

$$z = \frac{562 - 355}{69} = 3.00$$

This match sits exactly 3 standard deviations **above** the mean BPM. Under the empirical rule roughly 99.7% of matches fall within 3 standard deviations of the mean, so only about 0.15% of matches use this much boost or more - an extremely aggressive game.

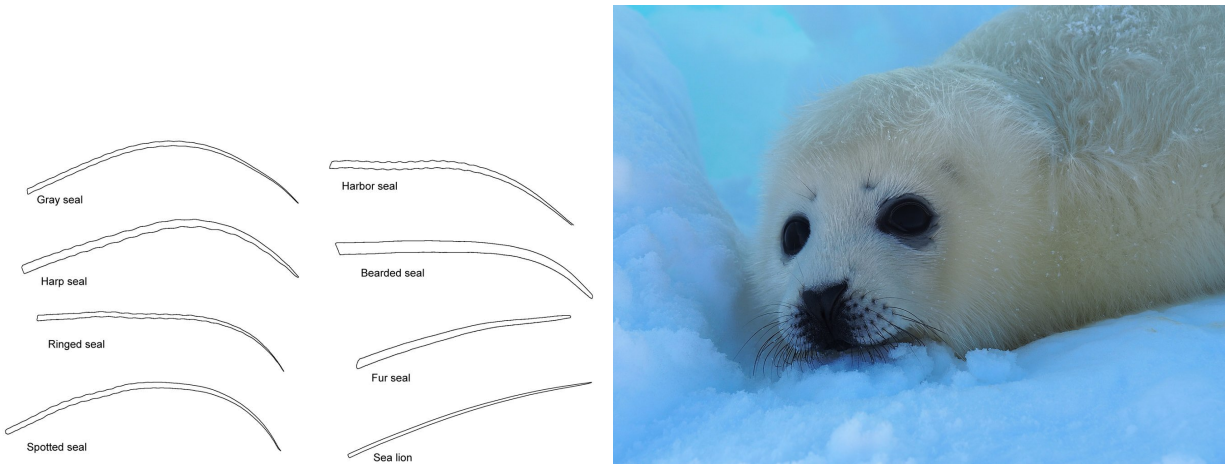


Figure 1: Pinniped whisker morphology - Ginter et al. 2012 (left), and a juvenile Harp Seal (right).

(**5**) Pinnipeds (seals and sea lions) possess the largest vibrissae (whiskers) among mammals, and their vibrissal hair shafts show a diversity of shapes (see **Figure 1**). In a study conducted by (Ginter et al. 2012), researchers measured 9 characteristics of Pinniped whiskers in individuals drawn from 9 species of seals and sea lions, with the goal of better characterizing whisker morphology and evolution. **Table 1** gives the recorded whisker lengths of 20 Harp Seals. The data have been sorted by increasing "Total Length (cm)" for your convenience. Use the table to answer the following questions.

Table 1: Whisker Length data for $n = 20$ Harp Seals

Observation	Whisker Total Length (cm)	Number of Beads
1	4.18	2
2	4.23	2
3	4.31	2
4	4.31	2
5	4.33	2
6	4.34	2
7	4.35	2
8	4.38	2
9	4.40	2
10	4.41	2
11	4.43	3
12	4.44	2
13	4.46	2
14	4.47	3
15	4.48	3
16	4.51	2
17	4.59	3
18	4.60	2
19	4.62	2
20	5.08	3

(a) Compute the quartiles, the interquartile range, the mean, and the standard deviation of the variable "Total Length (cm)". Use the values exactly as they appear in **Table 1**. Remember that we find the

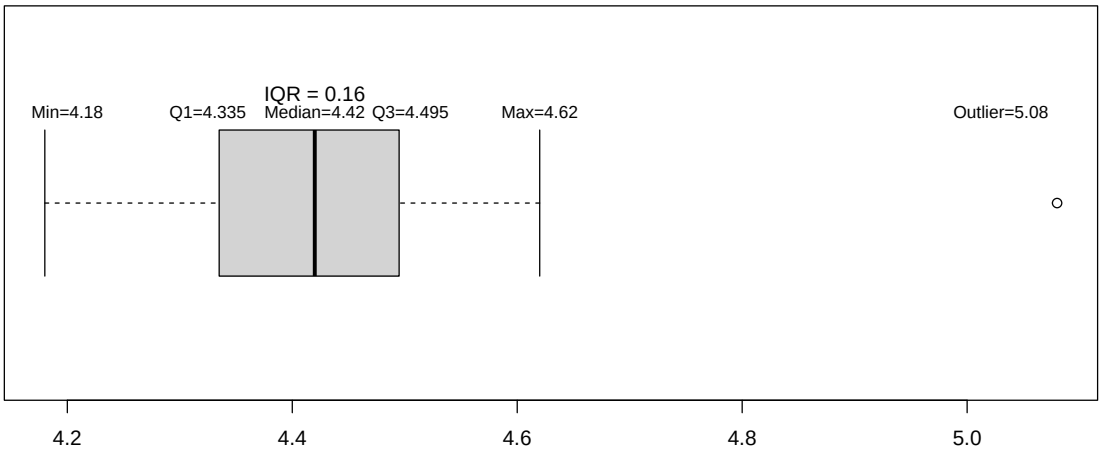
quartiles by the **median of halves** method used in class: find the median first, and when n is odd, leave that median out of both halves.

With $n = 20$ the median is the average of the 10th and 11th values, and each half holds 10 values, so $Q1$ is the average of the 5th and 6th and $Q3$ is the average of the 15th and 16th.

$$\begin{aligned} Q1 &= \frac{4.33 + 4.34}{2} = 4.335 \\ median &= \frac{4.41 + 4.43}{2} = 4.42 \\ Q3 &= \frac{4.48 + 4.51}{2} = 4.495 \\ IQR &= 4.495 - 4.335 = 0.16 \\ \bar{x} &= \frac{4.18 + 4.23 + \cdots + 5.08}{20} = 4.446 \\ s &= \sqrt{\frac{(4.18 - 4.446)^2 + (4.23 - 4.446)^2 + \cdots + (5.08 - 4.446)^2}{19}} = 0.189 \end{aligned} \tag{1}$$

(b) Construct a boxplot of the variable "Total Length" and note any outliers. Use the $1.5 \times IQR$ rule to decide what counts as an outlier.

The fences are $Q1 - 1.5 \times IQR = 4.335 - 0.24 = 4.095$ and $Q3 + 1.5 \times IQR = 4.495 + 0.24 = 4.735$. Only observation 20 (5.08 cm) falls outside them, so it is the single outlier.



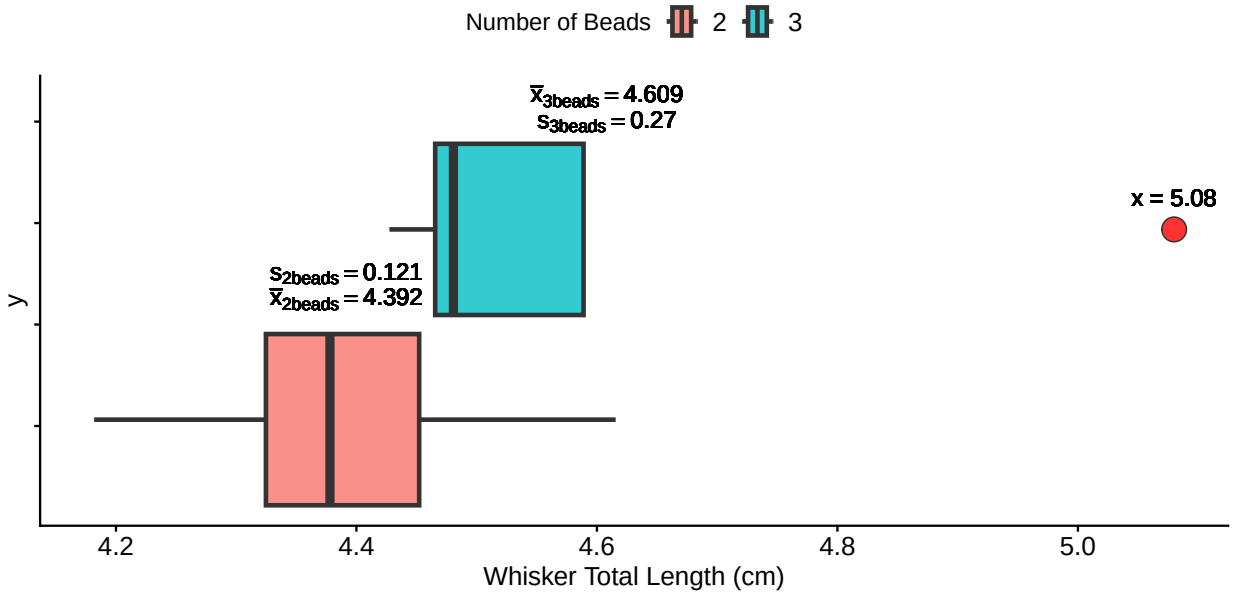
(c) Observation 20 has a total whisker length of 5.08 cm. How many standard deviations from the mean whisker length is this observation? (hint: compute the z-score for this observation)

The sample mean is $\bar{x} = 4.446$ and the sample standard deviation is $s = 0.189$, so

$$z = \frac{5.08 - \bar{x}}{s} = \frac{5.080 - 4.446}{0.189} = 3.35$$

Observation 20 lies about 3.35 standard deviations above the mean whisker length.

(**6**) The pair of boxplots below was constructed from the Pinniped whisker data in **Table 1**. The boxplots show the distribution of total whisker length (cm) for whiskers with 2 beads and for whiskers with 3 beads. Use them to answer the following questions.



(a) Using the boxplots above, do the whisker lengths for 2 bead whiskers and for 3 bead whiskers follow a normal distribution? Why or why not? Answer separately for each group.

The boxplot for 2 bead whiskers is roughly symmetric - the median sits near the middle of the box and the two whiskers are of similar length - which is consistent with, though not proof of, an approximately normal distribution.

The boxplot for 3 bead whiskers is clearly right-skewed. That group holds only $n = 5$ observations, and one of them (5.08 cm) sits far above the rest, dragging the upper whisker out. Those lengths are **not** approximately normal.

(b) Suppose researchers observe two new Harp Seals, denoted Seal 1 and Seal 2, and record the total whisker length and the number of beads for each. Seal 1 has a total whisker length of 4.32 cm on a whisker with 3 beads. Seal 2 has a total whisker length of 4.34 cm, but its whisker has only 2 beads. We want to know how the two compare *relative* to the distribution of whiskers with the same number of beads. Confirm that the whisker of Seal 1 has a z-score of -1.07 while the whisker of Seal 2 has a z-score of -0.43. Use the group means and standard deviations printed on the boxplot above.

Each seal must be standardized against the group with the **same** number of beads. From the figure, $\bar{x}_{2beads} = 4.392$ with $s_{2beads} = 0.121$, and $\bar{x}_{3beads} = 4.609$ with $s_{3beads} = 0.270$. Seal 1 has 3 beads, so

$$z_1 = \frac{4.32 - \bar{x}_3}{s_3} = \frac{4.32 - 4.609}{0.270} = -1.07$$

Seal 2 has 2 beads, so

$$z_2 = \frac{4.34 - \bar{x}_2}{s_2} = \frac{4.34 - 4.392}{0.121} = -0.43$$

Note that although Seal 2's whisker is the **longer** of the two in centimetres, Seal 1 sits further below its own group's mean. Comparing the raw lengths across the two groups would have given the opposite impression, and that is exactly what the z-score is for.

(c) Suppose the researchers observe a new whisker with 3 beads. Using the $\pm 2s$ rule, how long or how short would that whisker have to be before you would flag it as an outlier? Then look at the 3 bead group in **Table 1** and explain why the $\pm 2s$ rule should be treated with caution here. (hint: your answer to part (a) for the 3 bead group is the reason, and note that the $1.5 \times IQR$ rule and the $\pm 2s$ rule do not always flag the same observations)

Under the $\pm 2s$ rule an observation is flagged when it is at least two standard deviations from the mean, $x \le \bar{x} - 2s$ **or** $x \ge \bar{x} + 2s$, so the whisker would have to be at least

$$\bar{x}_3 + 2s_3 = 4.609 + 2 \times 0.270 = 5.149 \text{ (cm)}$$

long, or no longer than

$$\bar{x}_3 - 2s_3 = 4.609 - 2 \times 0.270 = 4.069 \text{ (cm)}$$

Why to be cautious: the $\pm 2s$ rule is built on the empirical rule, which assumes an approximately normal distribution - and in part (a) we decided the 3 bead group is not normal. Worse, $\bar{x}_3$ and s_3 are themselves computed from only five observations, one of which (5.08) is the extreme value we might want to detect. That single point inflates s_3 and pushes the upper cut-off out to 5.149, so the rule does **not** flag 5.08 at all - even though the $1.5 \times IQR$ rule applied to the full sample in Problem 5(b) did flag it. The two rules can disagree, and neither should be applied mechanically to a small, skewed group.

(**7**) According to the World Health Organization, the heights of girls aged five years are approximately normal with a mean of $\mu = 109$ cm and a standard deviation of $\sigma = 4.5$ cm. Use the empirical rule to answer parts (a) through (e). No table or calculator is needed — every number below works out evenly.

(a) Give the interval, centred on the mean, that contains approximately 68% of the heights. Then do the same for 95% and for 99.7%.

Each interval is the mean plus or minus a whole number of standard deviations.

- 68%: $\mu \pm 1\sigma = 109 \pm 4.5$, which is **104.5 cm to 113.5 cm**.
- 95%: $\mu \pm 2\sigma = 109 \pm 9$, which is **100 cm to 118 cm**.
- 99.7%: $\mu \pm 3\sigma = 109 \pm 13.5$, which is **95.5 cm to 122.5 cm**.

(b) Approximately what percentage of five-year-old girls are between 95.5 cm and 122.5 cm tall?

Those are exactly the endpoints of the third interval in part (a): $95.5 = \mu - 3\sigma$ and $122.5 = \mu + 3\sigma$. By the empirical rule, approximately 99.7% of heights fall there — almost all of them.

(c) A five-year-old girl is 100 cm tall. Compute her z-score and interpret it in one sentence.

$$z = \frac{100 - 109}{4.5} = \frac{-9}{4.5} = -2$$

She is **two standard deviations below** the mean height for her age. The negative sign is what says *below*; the size, 2, is measured in standard deviations rather than centimetres.

(d) Approximately what percentage of five-year-old girls are **taller** than 113.5 cm? (hint: start from your 68% interval in part (a), and remember the distribution is symmetric)

$113.5 = \mu + 1\sigma$, so this is the upper tail beyond one standard deviation. The empirical rule puts 68% *inside* $\mu \pm 1\sigma$, which leaves $100\% - 68\% = 32\%$ outside it. A normal distribution is symmetric, so that 32% splits evenly between the two tails:

$$\frac{32\%}{2} = 16\%$$

Approximately 16% of five-year-old girls are taller than 113.5 cm.

(e) Would a height of 100 cm be flagged as an outlier under the $\pm 2s$ rule? Explain.

Yes, just. From part (c) its z-score is -2, and the $\pm 2s$ rule from class flags any value **at least** two standard deviations from the mean, $|z| \ge 2$. A height of 100 cm sits exactly on that boundary, so it is flagged as a (low) outlier.

It is worth seeing how unusual that is: the 95% interval from part (a) leaves 5% outside, split evenly, so only about 2.5% of five-year-old girls are 100 cm tall or shorter.

Part III: Probability and Discrete Random Variables

(**8**) Define each of the following terms and concepts.

(a) Sample space

The set of all possible outcomes of a random trial.

(b) Event

A subset of the sample space; a collection of one or more possible outcomes.

(c) Random trial

A process or experiment whose individual outcome cannot be predicted in advance, but for which the set of possible outcomes is known.

(d) Union of two events

The event $A \cup B$ that A occurs, or B occurs, or both.

(e) Intersection of two events

The event $A \cap B$ that A **and** B both occur.

(f) Disjoint (mutually exclusive) events

Two events that cannot both occur on the same trial, so $P(A \cap B) = 0$.

(g) Random variable

A variable whose numerical value is determined by the outcome of a random trial.

(h) Discrete random variable

A random variable that can take on only a countable set of values, often whole numbers, such as the number of heads in three coin flips.

(i) Continuous random variable

A random variable that can take on any value in an interval, such as a height or a waiting time.

(j) Probability distribution

A listing or rule that gives every possible value of a random variable together with its probability. The probabilities each lie between 0 and 1 and sum to 1.

(k) The Law of Large Numbers

As the number of trials increases, the relative frequency with which an event occurs converges to the true probability of that event.

(**9**) Answer the following probability questions, and show your work.

(a) A fair coin is flipped 3 times. Show how to compute the probability that at least one head is observed. (hint: it is easier to find the probability of the complement first)

The complement of "at least one head" is "no heads at all", that is, three tails:

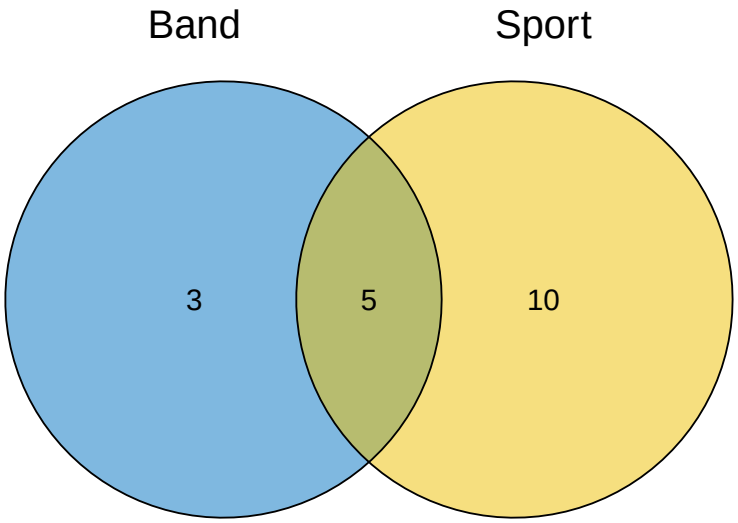
$$P(\text{no heads}) = 0.5 \times 0.5 \times 0.5 = 0.125$$

By the complement rule,

$$P(\text{at least one head}) = 1 - 0.125 = 0.875$$

(b) In a class of 30 students, 8 students are in band, 15 students play a sport, and 5 students are both in band and play a sport. Let A be the event that a student is in band and let B be the event that a student plays a sport. Draw a Venn diagram that models this situation, filling in the number of students in each region, then show how to compute the probability that a student is in band **or** plays a sport.

The Venn diagram has $8 - 5 = 3$ students in band only, 5 in the overlap, $15 - 5 = 10$ in sport only, and $30 - (3 + 5 + 10) = 12$ outside both circles.



Because the two events are **not** disjoint - 5 students are in both - we must subtract the overlap so that it is not counted twice. This is the addition rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{8}{30} + \frac{15}{30} - \frac{5}{30} = \frac{18}{30} = 0.60$$

(c) A standard deck of 52 cards contains four suits (Hearts, Diamonds, Clubs, and Spades). Each suit contains 13 cards: 9 numbered cards, 3 face cards, and an Ace. A single card is drawn at random. Show how to compute the probability that it is a numbered card, and the probability that it is the Ace of Spades.

There are $4 \times 9 = 36$ numbered cards, so

$$P(\text{numbered}) = \frac{36}{52} = 0.692$$

There is exactly one Ace of Spades, so

$$P(\text{Ace of Spades}) = \frac{1}{52} = 0.019$$

(d) An urn contains 5 green balls, 7 red balls, and 4 yellow balls. Two balls are selected at random **with replacement**. Show how to compute the probability that the first ball is yellow and the second ball is red.

There are $5 + 7 + 4 = 16$ balls. With replacement the two draws are independent, so by the multiplication rule

$$P(\text{yellow, then red}) = \frac{4}{16} \times \frac{7}{16} = \frac{28}{256} = 0.109$$

(e) Repeat part (d), but now assume the two balls are selected **without replacement**. Explain why your answer changed.

Without replacement the first ball is not returned, so only 15 balls remain for the second draw, and all 7 red balls are still among them:

$$P(\text{yellow, then red}) = \frac{4}{16} \times \frac{7}{15} = \frac{28}{240} = 0.117$$

The answer changed because the draws are no longer independent - the outcome of the first draw changes the composition of the urn, and therefore the probability of the second. Here, removing a yellow ball leaves red slightly more likely, so the probability rises.

(f) On a given day in Moscow, Idaho, the probability that it rains is $P(R) = 0.1$, the probability that it is windy is $P(W) = 0.2$, and the probability that it is cloudy is $P(C) = 0.7$. Assume these three weather events are independent. Show how to compute the probability that a given day is rainy, cloudy, **and** windy.

For independent events the multiplication rule extends to all three:

$$P(R \cap C \cap W) = 0.1 \times 0.7 \times 0.2 = 0.014$$

(**10**) A single fair coin is flipped once. Let $X = 1$ if the flip comes up heads and $X = 0$ if it comes up tails. A random variable with only these two outcomes is called a **Bernoulli** random variable, and its probability of success is $p = P(X = 1)$.

(a) Write down the probability distribution of X as a table of values and their probabilities.

For a fair coin $p = 0.5$, so

x	$P(x)$
0	0.5
1	0.5

The two probabilities are each between 0 and 1 and sum to 1, so this is a valid probability distribution.

(b) Show how to compute the mean of X using $\mu = \sum_x x \cdot P(x)$, and confirm that it equals p.

$$\mu = 0(0.5) + 1(0.5) = 0 + 0.5 = 0.5$$

which is exactly p. This is worth pausing on: the mean of a Bernoulli variable is **always** p, because the only term that survives the sum is the one where $x = 1$.

(c) Show how to compute the variance of X using $\sigma^2 = \sum_x (x - \mu)^2 \cdot P(x)$, and confirm that it equals $p(1 - p)$.

$$\sigma^2 = (0 - 0.5)^2(0.5) + (1 - 0.5)^2(0.5) = 0.25(0.5) + 0.25(0.5) = 0.125 + 0.125 = 0.25$$

and $p(1 - p) = 0.5(0.5) = 0.25$, which agrees.

(d) Now suppose the coin is bent so that it comes up heads only 30% of the time. Show how to compute the mean and the variance of X for this bent coin.

Now $p = 0.3$, so

$$\mu = 0(0.7) + 1(0.3) = 0.3 \qquad \sigma^2 = p(1 - p) = 0.3(0.7) = 0.21$$

Note that the variance went **down** from 0.25. A Bernoulli variable is most variable when $p = 0.5$, because that is when the outcome is hardest to predict; a coin that almost always lands tails is much more predictable.

(e) A **binomial** random variable counts the number of successes in n independent Bernoulli trials. If the fair coin from parts (a)-(c) is flipped 10 times, show how to compute the mean and the variance of the number of heads. (hint: $E[X] = np$ and $Var[X] = np(1 - p)$)

With $n = 10$ and $p = 0.5$,

$$E[X] = np = 10(0.5) = 5 \text{ heads} \qquad Var[X] = np(1-p) = 10(0.5)(0.5) = 2.5$$

Both are just the Bernoulli answers from (b) and (c) multiplied by n — which is the whole point: a binomial variable is the sum of n independent Bernoulli trials, so its mean and variance are n times theirs.

(**11**) Consider a survey of tree volume. Suppose that in the population of trees, 10% have a volume of 10 cubic feet, 20% have a volume of 20 cubic feet, 30% have a volume of 30 cubic feet, and 40% have a volume of 40 cubic feet. Let X denote the volume of a randomly selected tree, and assume every tree has an equal chance of being selected. The probability distribution of X is given in **Table 2** below.

Table 3: Probability distribution of tree volume X

Tree Volume (cubic feet)	$P(X)$
10	0.1
20	0.2
30	0.3
40	0.4

(a) Verify that the four probabilities in **Table 2** form a valid probability distribution.

Each probability lies between 0 and 1, and they sum to one: $0.1 + 0.2 + 0.3 + 0.4 = 1.0$. Both conditions hold, so this is a valid probability distribution.

(b) Show how to compute the mean of X using $\mu = \sum_x x \cdot P(x)$, and confirm that it equals $\mu = 30$ cubic feet.

$$\mu = 10(0.1) + 20(0.2) + 30(0.3) + 40(0.4) = 1 + 4 + 9 + 16 = 30 \text{ cubic feet}$$

(c) Show how to compute the variance of X using $\sigma^2 = \sum_x (x-\mu)^2 \cdot P(x)$, then take the square root and confirm that the standard deviation is $\sigma = 10$ cubic feet.

$$\begin{aligned} \sigma^2 &= (10-30)^2(0.1) + (20-30)^2(0.2) + (30-30)^2(0.3) + (40-30)^2(0.4) \\ &= 400(0.1) + 100(0.2) + 0(0.3) + 100(0.4) \\ &= 40 + 20 + 0 + 40 \;=\; 100 \end{aligned} \tag{2}$$

$$\sigma = \sqrt{100} = 10 \text{ cubic feet}$$

(d) Is X, as described here, a discrete or a continuous random variable? Explain.

As described here X is **discrete**: it takes only the four listed values, and each has its own probability. Tree volume measured in the real world would be continuous, since any value in an interval is possible, but this population has been simplified to four volume categories.

(**12**) Answer the following, showing the formula you used and the numbers you substituted into it.

(a) An angler has a 40% chance of catching a fish on every cast, and each cast is independent of the others. Using the binomial probability mass function

$$P(X = k) = \frac{n!}{k!(n-k)!} \cdot p^k \cdot (1-p)^{n-k}$$

show how to compute the probability that she catches exactly 2 fish in her next 5 casts.

Here $n = 5$, $k = 2$, and $p = 0.4$:

$$P(X = 2) = \frac{5!}{2!(5-2)!}(0.4)^2(0.6)^3 = 10 \times 0.16 \times 0.216 = 0.3456$$

So there is about a 34.6% chance that she catches exactly two fish.

(b) For the angler in part (a), show how to compute the mean number of fish she expects to catch in 5 casts, and the variance. (hint: $E[X] = np$ and $Var[X] = np(1-p)$)

$$E[X] = np = 5(0.4) = 2 \text{ fish} \qquad Var[X] = np(1-p) = 5(0.4)(0.6) = 1.2$$

(c) A small campus coffee stand serves an average of 3 customers per 10-minute period. Let X be the number of customers in a 10-minute period, and assume X follows a Poisson distribution with $\lambda = 3$. Using

$$P(X = k) = \frac{\lambda^k \cdot e^{-\lambda}}{k!}$$

show how to compute the probability that **no** customers arrive in a given 10-minute period.

With $\lambda = 3$ and $k = 0$, and recalling that $0! = 1$ and $3^0 = 1$:

$$P(X = 0) = \frac{3^0 e^{-3}}{0!} = e^{-3} = 0.0498$$

So there is about a 5% chance of no customers.

(d) For the coffee stand in part (c), show how to compute the probability that **at most one** customer arrives in a given 10-minute period.

"At most one" means $k = 0$ or $k = 1$, so we add the two probabilities:

$$P(X = 1) = \frac{3^1 e^{-3}}{1!} = 3e^{-3} = 0.1494$$

$$P(X \leq 1) = P(X = 0) + P(X = 1) = 0.0498 + 0.1494 = 0.1992$$

So there is about a 20% chance of one customer or fewer.

Part IV: Standard Error and the Margin of Error

(**13**) A population has a mean of $\mu = 75$ and a standard deviation of $\sigma = 12$. You repeatedly draw random samples of $n = 36$ observations from this population, and each time you record the sample mean $\bar{x}$.

(a) Show how to compute the **standard error** of the sample mean, $SE = \sigma/\sqrt{n}$.

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{36}} = \frac{12}{6} = 2$$

(b) A classmate insists that the standard error should be 12, because that is the standard deviation of the population. Explain the mistake in their reasoning.

They have confused the spread of **individual observations** with the spread of the **sample mean**. The value $\sigma = 12$ describes how far a single observation typically falls from μ. The standard error describes how far a whole sample's *average* typically falls from μ — and an average is far more stable than one observation, because high and low values within the sample cancel each other out.

Dividing by $\sqrt{n}$ is exactly what accounts for that cancelling. The two are equal only when $n = 1$, where the "sample mean" is just a single observation.

(c) Using the $\pm 2 \cdot SE$ rule from class, show how to compute the **margin of error** for $\bar{x}$, and give the interval it defines around μ.

$$\text{margin of error} = 2 \cdot SE = 2(2) = 4$$

so the interval is

$$\mu \pm 4 = 75 \pm 4, \text{ that is, from 71 to 79}$$

About 95% of the sample means we could draw will land in that interval.

(d) Show how to compute the standard error if the sample size is increased to $n = 144$. What happens to the margin of error, and why?

$$SE = \frac{12}{\sqrt{144}} = \frac{12}{12} = 1$$

so the margin of error becomes $2(1) = 2$, and the interval narrows to 75 ± 2, from 73 to 77.

The sample size was multiplied by 4, and the margin of error was **halved**. That is because the standard error depends on $\sqrt{n}$, not on n: $\sqrt{4} = 2$, so quadrupling the sample size cuts the error in half. Buying twice the precision costs four times the data, which is why large surveys get expensive quickly.

References

Ginter, Carly C, Thomas J DeWitt, Frank E Fish, and Christopher D Marshall. 2012. "Fused Traditional and Geometric Morphometrics Demonstrate Pinniped Whisker Diversity." *PloS One* 7 (4): e34481.